Cristian Curaba

Udine – Italy
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About me

A passionate mathematician and coder dedicated to driving impactful research through technical expertise. With an eclectic, creative approach, I excel in collaborative environments, prioritizing trustworthiness, and intellectual rigor. Active in AI safety community building (AIS4I, PauseAI Italia) and in the resilience of essential services.

Education

Master's Degree at University of Trieste, Italy
Data Science and Scientific Computing, Grade: 110/110 cum Laude *2022 – 2024*
Thesis: *Integrating Large Language Models and Formal Verification for Automated Cryptographic Protocol Vulnerability Detection*

School for Advanced Studies of Udine – Second-level Master's degree
Student of the Scientific Class, Grade: 110/110 *2019 – 2024*
The School for Advanced Studies of the University of Udine "Di Toppo Wassermann" is a higher learning institution based on merit. After a selective admission, provides a five-year scholarship covering tuition, board, and lodging. During this period, students have to attend extra courses and exams, culminating in a second-level Master's degree diploma. Learn more at https://superiore.uniud.it/en.

Bachelor's Degree at University of Udine, Italy
Mathematics, Grade: 110/110 *2019 – 2022*
Thesis: *Teoria descrittiva della complessità: la logica FP+C cattura Ptime nella classe dei grafi ad intervallo*

Work Experience

University of Udine
Research Fellowship *2025 – Now*
Development and implementation of advanced methods to support decision-making in emergency scenarios involving critical infrastructure.
Supervisor: Prof. Stefano Grimaz - Chairholder of the UNESCO Chair in Intersectoral Safety

Apart Fellowship
AI Safety Research *2024 (6 months)*
Developed **CryptoFormalEval**: a benchmark aimed at assessing the capability of LLMs to detect and analyze vulnerabilities in cryptographic protocols.

SMS Group
Development and Research *2023 – 2024 (4 months)*
Applied machine learning methods and data analysis to devise strategies for controlling electric furnaces.

Computer and Programming Skills

Advanced: C, LaTeX, C++, Python (PyTorch)
Intermediate: R, Shell Unix, PostgreSQL, Git
Basic: Pyro, OpenMP, MPI, HTML, Matlab, Tamarin

Languages

Italian: Mother tongue

English: Proficient user (C1) *IELTS*

Experiences

- UNESCO Science Communication Training (2026).
  Participated in the UNESCO Science Communication Training for Young Professionals and Early Career Researchers.
- AIS4I (2025 – Now).
  Active member, contributing to technical and didactic working groups. Website link
- PauseAI Italia (2025 – Now).
  Supporting PauseAI Italia's advocacy and outreach work.
- AI Safety Udine (2025 – Now).
  Building a local AI safety group in Udine.
- Introduction to AGI Risks - Student Course (2026).
  Designed and led an informal introductory course on AGI risks for students at University of Udine.
- AGI Discussion Group (2025).
  Leading a discussion group on AGI topics at University of Udine for researchers and major students.
- AI Act Summer School (2024).
  Participated in the AI Act Summer School at University of Udine, a 3-day program on AI ethics, AI policy, and AI governance.
- Alignment Mapping Program (2024).
  Successfully completed the Alignment Mapping Program, an 8-week program on AGI issues, AGI plans, and personal career development.
- ML4Good French (2024).
  Participation in the 10-day bootcamp program on Machine Learning for AGI Safety run by EffiSciences.
- Testing Language Models for Autonomous Capabilities - Apart Hackathon.
  Winner as best Quality Assurance Tester. Evaluated Task: Tamarin Formalizer.
- AI Safety North course (2023).
  Fundamentals of Artificial General Intelligence (about 40 hours). Shared insights in a weekly-planned discussion group.
- International Collegiate Programming Contest (SWERC - 2023/2024).
  Participation in the international programming competition (C++) in the team of The School for Advanced Studies "Di Toppo Wassermann". Classified 55/103.
- Sisifo Association (2020 – 2024).
  Actively participating in a non-profit association focused on raising awareness about criminal law, the rehabilitative function of punishment, and related topics. Facebook page
- Interpelliamo.it (2026).
  A free, transparent platform collecting the "interpelli" published on Italian state schools' online boards, gathered twice a day and filterable by province and teaching class, with email alerts. Website link

Publications

[1]: **Cristian Curaba** and Denis D'Ambrosi and Alessandro Minisini and Natalia Pérez-Campanero Antolín. *CryptoFormalEval: Integrating LLMs and Formal Verification for Automated Cryptographic*

Protocol Vulnerability Detection.
NeurIPS 2024 Workshop: Sys2-Reasoning Poster
ArXiv: arXiv:2411.13627

[2]: A. Brunello, **Cristian Curaba**, L. Geatti, M. Mignani, A. Montanari, N. Saccomanno. *Fixing FOLIO and MALLS: Verified Annotations and an LLM-assisted Framework to Focus Human Relabeling*.
ArXiv: arXiv:2606.02837

[3]: **Cristian Curaba** and S. Grimaz. *Series and Parallel Dependency Links: Vitality for Interdependent Networks of Networks*.
Complex Networks 2026
ArXiv: arXiv:2610.21843

[4]: **Cristian Curaba** and S. Grimaz. *Composable, Knowledge-Driven Disservice Propagation for Complex Interdependent Services with a Flow-Allocation Module Validated Against Hydraulic Simulation*.
Complex Networks 2026
ArXiv: arXiv:2610.26417

[5]: A. Brunello, **Cristian Curaba**, L. Geatti, M. Mignani, A. Montanari, N. Saccomanno. *Natural Language to First-Order Logic LLM-based Autoformalization*.
EMNLP 2026
ArXiv: arXiv:2610.12030

Projects

CryptoFormalEval: Introduced a benchmark for testing how well LLMs can find vulnerabilities in cryptographic protocols. By combining LLMs with symbolic reasoning tools like Tamarin, we aim to improve the efficiency and thoroughness of protocol analysis, paving the way for future AI-powered cybersecurity defenses.
Link: https://github.com/Cristian-Curaba/CryptoFormalEval

CASCADE: A platform for modelling interdependent critical infrastructure (power, water, heat, digital, personnel) as a graph, then simulating how a failure in one element propagates through the network via a rule-driven, category-typed propagation engine: quantity-aware and purely logical interdependencies share one discrete functionality scale and converge through a single monotonic update rule.
Platform: https://cascade-platform.org/
Code: https://github.com/cascade-platform-org/CASCADE

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